

3.7.4 External hardware devices

3.7.4.1 Input and output devices

Content	Additional information
Know the main characteristics, purposes and suitability of the devices and understand their principles of operation.	Devices that need to be considered are: • barcode reader • digital camera • laser printer • RFID.

3.7.4.2 Secondary storage devices

Content	Additional information
Explain the need for secondary storage within a computer system.	
Know the main characteristics, purposes, suitability and understand the principles of operation of the following devices: • hard disk • optical disk • solid-state disk (SSD).	SSD = NAND flash memory + a controller that manages pages, and blocks and complexities of writing. Based on floating gate transistors that trap and store charge. A block, made up of many pages, cannot overwrite pages, page has to be erased before it can be written to but technology requires the whole block to be erased. Lower latency and faster transfer speeds than a magnetic disk drive.
Compare the capacity and speed of access of various media and make a judgement about their suitability for different applications.	